CENTRE FOR ASTROPHYSICS & SUPERCOMPUTING

Upgrading COSMOS

Making the SAO Encyclopedia of Astronomy the world's premier interactive astronomy reference — a proposal for the next generation of COSMOS.

Swinburne University of Technology · March 2026

CURRENT STATE

COSMOS Today

A respected, free astronomy encyclopedia — recently modernised with Swinburne house styling and fully self-hosted infrastructure.

644	866	20+	4
articles	images	years of content	major citing platforms

Cited by Universe Today · Astronomy Cast · Astronomy Picture of the Day · timeanddate.com

LANDSCAPE

The Competition

How COSMOS compares to other online astronomy resources.

RESOURCE	ARTICLES	DEPTH	AUTHORITY	INTERACTIVE	FREE
COSMOS	644	High	Research astronomers	Limited	Yes
Wikipedia	30,000+	Variable	Crowd-sourced	No	Yes
NASA (fragmented)	1,000s	Moderate	NASA	Some	Yes
HyperPhysics	~6,000	Moderate	Academic	Concept maps	Yes
Britannica	~500-800	Moderate	Editors	Quizzes	Paywalled
ESA/Hubble	41	Low	ESA	No	Yes

POSITIONING

What Makes COSMOS Unique

No other resource combines all four of these qualities.

Expert-Written

Every article written by active research astronomers at Swinburne CAS. Unlike Wikipedia (crowd-sourced) or Britannica (generalist editors).

Free & Open

No paywall, no login, no ads. Unlike Britannica (paywalled) or NASA (fragmented across dozens of sub-sites).

Astronomy-Focused

Dedicated astronomy encyclopedia with real depth. Unlike general encyclopedias where astronomy is one section among hundreds.

Full Control

We own the entire site — code, content, hosting. Can add any feature without committee approval or platform constraints.

The Gap Nobody Fills

No astronomy resource offers **interactive visualisations** tied to encyclopedic content. No one provides a **navigable knowledge graph**. No one bridges **textbook definitions** with **current research**. No one offers **structured data** for apps and LLMs to query.

These are all achievable with the infrastructure we now have.

UPGRADE #1 · VERY HIGH IMPACT

Interactive Visualisations

Replace static images with explorable 3D diagrams. A prototype already exists for the Zenith article.

Flagship Targets

- **HR Diagram** — hover stars, see properties, click to articles
- **Galaxy Classification** — explore the Hubble tuning fork
- **Cosmic Distance Ladder** — step through each rung interactively
- **Celestial Sphere** — adjustable latitude, rotating sky
- **EM Spectrum** — slide through wavelengths, see telescopes
- **Stellar Evolution** — animate a star's life cycle

Technical Approach

- Three.js (3D) and D3.js (data viz) — no build step, CDN-loaded
- Self-contained HTML+JS per visualisation
- Embeddable in article pages or standalone
- Mobile-responsive, touch-friendly
- Each diagram links back to relevant COSMOS articles

Prototype: experimental/zenith-interactive.html

UPGRADE #2 · HIGH IMPACT, LOW EFFORT

Knowledge Graph Explorer

COSMOS already has 644 densely cross-linked articles. Visualise the connections.

HyperPhysics pioneered concept-map navigation in 1998 — but with crude bubble diagrams frozen in time. We can build a modern, interactive force-directed graph where users visually explore how concepts connect.

Click "Black Hole" and see its connections radiate outward: Event Horizon, Singularity, Hawking Radiation, Neutron Star, Supernova, General Relativity...

Why This Works

- Data already exists — just parse existing cross-links
- Single page with D3.js force-directed layout
- Click any node to read the article
- Filter by letter, topic cluster, or search
- Visually demonstrates the depth and interconnectedness of COSMOS

UPGRADE #3 · HIGH IMPACT, LOW EFFORT

Structured Data & SEO

Make COSMOS machine-readable. Immediate SEO benefit, LLM integration, and app ecosystem.

JSON-LD Markup

Add Schema.org Article markup to every page. Enables Google Knowledge Panels and rich search results. The single highest-ROI SEO action available.

Cross-Link Graph

Export the article relationship graph as JSON. Apps, LLMs, and researchers can query how astronomy concepts connect — no other encyclopedia offers this.

Social Cards

OpenGraph and Twitter Card meta tags for every article. When someone shares a COSMOS link, it displays with title, description, and image — driving engagement.

UPGRADE #4 · HIGH IMPACT

Bridge Definitions & Current Research

Nobody links canonical astronomy definitions with the latest papers. COSMOS + NASA ADS can.

Each article gets a "Recent Research" sidebar, auto-populated from NASA ADS. A GitHub Actions workflow queries ADS weekly for papers matching each article's keywords.

The article on "Dark Energy" would show the 5 most recent relevant papers, with titles, authors, and links. No manual maintenance needed.

Why This Matters

- Positions COSMOS as a living, current resource
- Connects undergraduate learning with the research frontier
- Auto-updated — no editorial burden
- Unique: Astrobites covers daily papers but doesn't link to definitions. COSMOS would do both.

ROADMAP

Phased Implementation

Phase 1: Quick Wins

Weeks

- JSON-LD schema markup
- OpenGraph / Twitter cards
- PWA offline capability
- Knowledge graph data extraction

Phase 2: Showcase

1-2 months

- 5 flagship interactive visualisations
- Knowledge graph explorer page
- Social media sharing improvements
- Embed support for educators

Phase 3: Sustained

Ongoing

- ADS research sidebar
- Multi-level reading depth
- New article contributions
- Multilingual expansion

PROGRESS

What's Already Built

The foundation is in place. The entire COSMOS site has been modernised and migrated.

Completed

- All 644 articles transferred to modern static site
- 866 images downloaded and served locally
- Swinburne house style applied (responsive, mobile-friendly)
- Client-side search and dynamic letter browsing
- GitHub Pages hosting with auto-deploy on push
- Zero dependencies on old astronomy.swin.edu.au server
- Interactive zenith prototype built (Three.js)
- Transfer automation and verification tooling

Infrastructure

- **Repo:** github.com/ivolabbe/cosmos
- **Live:** ivolabbe.github.io/cosmos
- Static HTML — fast, secure, no server maintenance
- GitHub Actions CI/CD — push to deploy
- Full transfer pipeline documented and automated
- Verification script catches broken links/images
- Agent architecture for parallel batch operations

Let's Make COSMOS the World's Best Astronomy Reference

Expert-written. Free. Interactive. Modern.

No other resource can claim all four.

Live site: ivolabbe.github.io/cosmos

Repository: github.com/ivolabbe/cosmos